



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VNC GAS STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

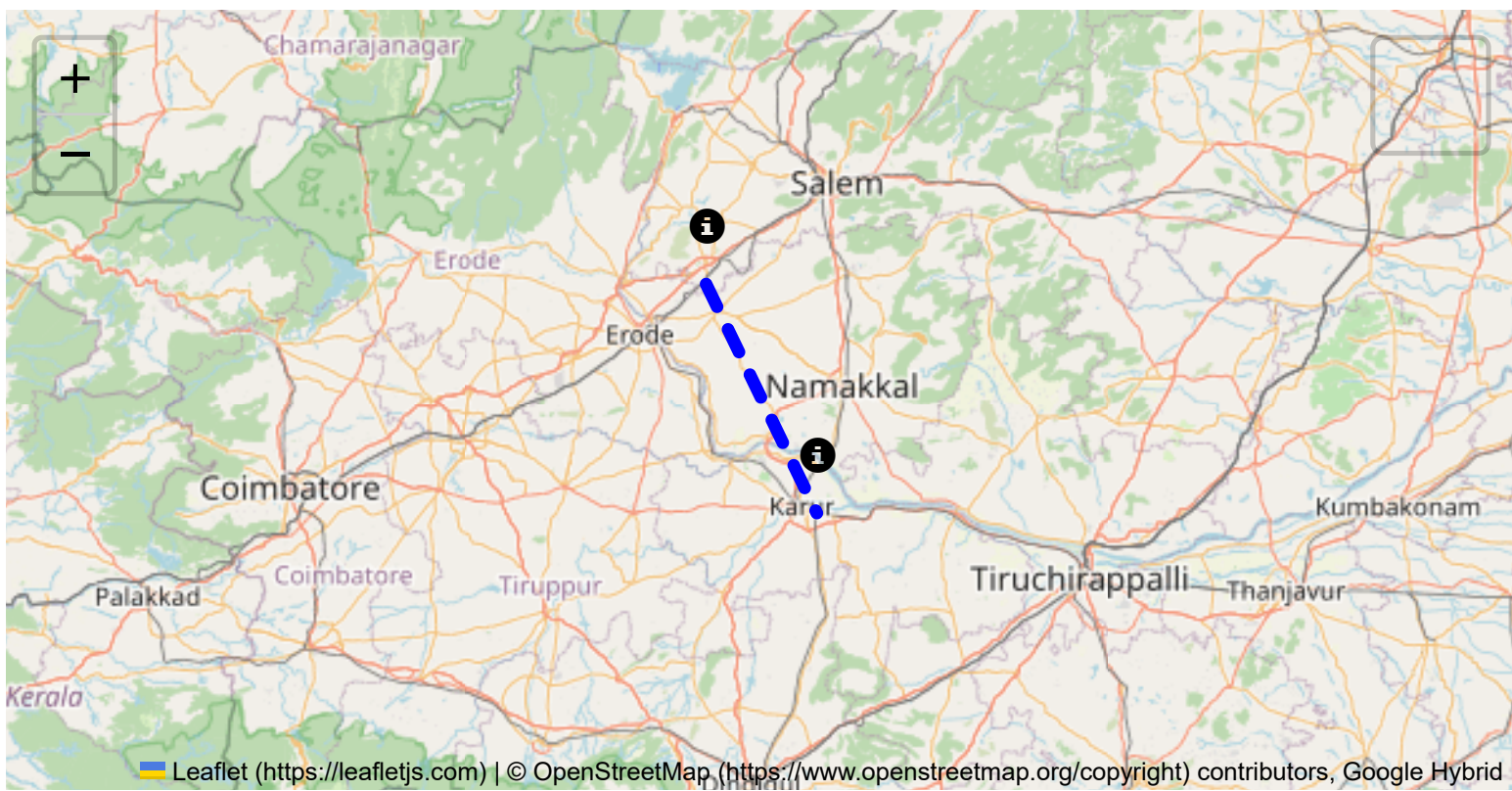
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

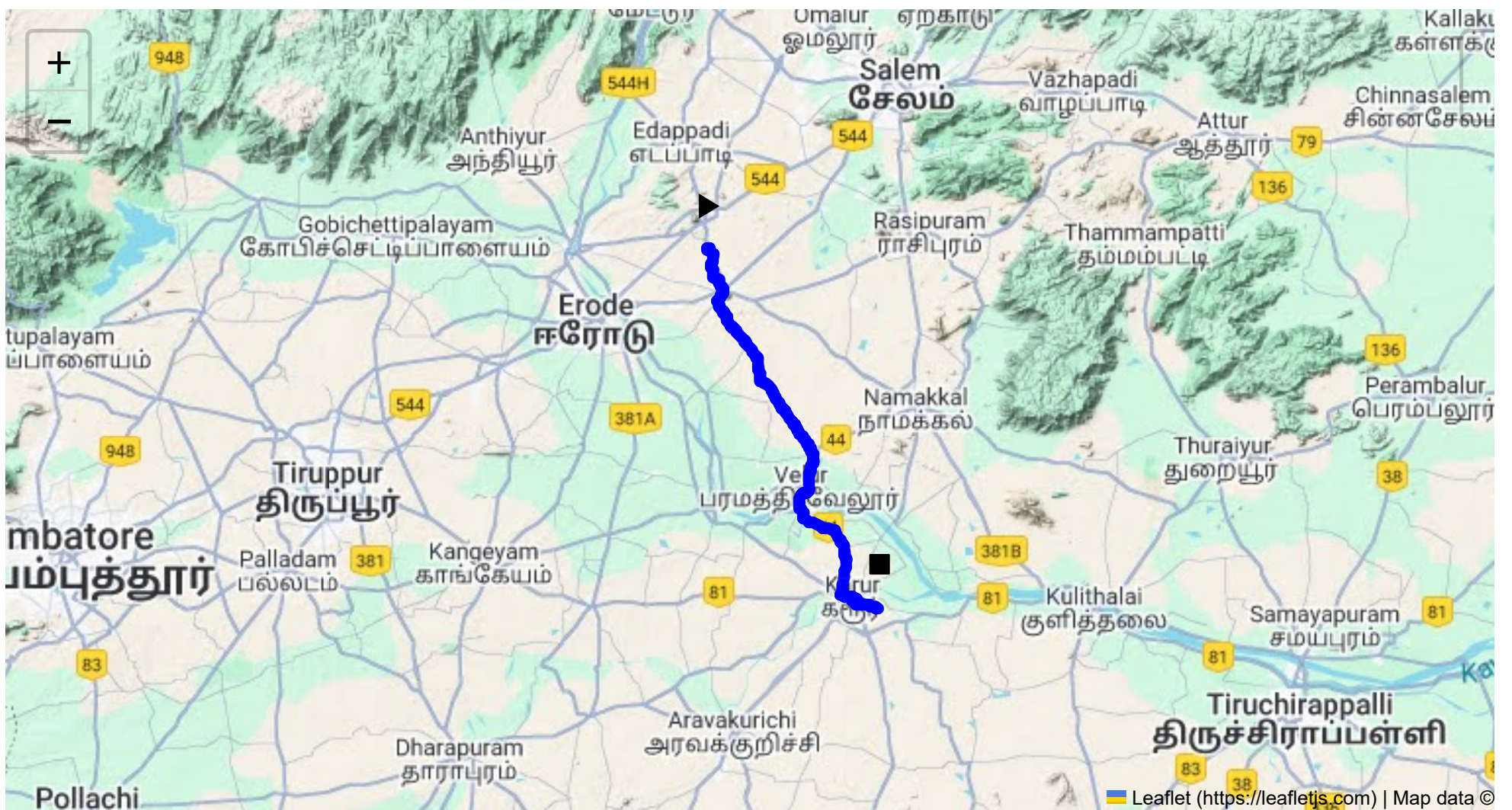
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 71.00 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (10.94472, 78.111602)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, to No. B-35, Trichy Main Road, covers approximately 71 kilometers. The journey mostly follows arterial roads that connect various towns and rural areas. The key highways include NH44 and NH81, which facilitate relatively smooth travel between these locations.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with significant seasonal variations:

- **Summer (March to June):** High temperatures that can reach up to 40°C, which can cause road surface issues like melting tar.
- **Monsoon (July to September):** Heavy rainfall can lead to flooding, reducing road visibility and traction.
- **Winter (October to February):** Moderates in temperatures, usually benign but early morning fog can reduce visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, traffic congestion occurs during morning hours (8-10 AM) and evening rush hours (5-7 PM).
- **Congestion-Prone Areas:** Town centers and junctions close to the Karur district are known for heavy traffic. Sangagiri's local markets can also slow down traffic.

4. Assessment of Road Quality and Infrastructure

- **NH44 & NH81** are generally well-maintained, but vigilance is needed near intersections and town entries where road conditions can degrade.
- Rural roads may experience potholes and uneven surfaces, especially post-monsoon.

5. Suggestions for Alternative Routes for Emergencies

- If NH44 is congested, consider using smaller state highways that run parallel to major sections.
- For detours, local roads connecting to district centers can be used; however, always verify current conditions as they can be unpredictable.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Permits are required for transporting hazardous materials.
- Transport hours may be restricted; night travel is usually preferable for hazardous materials to avoid peak traffic.
- Ensure materials are labeled and secured according to Indian transport regulations.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Accidents are more frequent post-monsoon due to slippery roads. Incidents involving hazardous materials are rare but often occur due to human error or poor road conditions.

8. Environmental Considerations and Sensitive Areas

- **Water Bodies:** Be cautious near the Kaveri River and its tributaries to avoid contamination risks.
- **Wildlife Crossings:** There are few, but drivers should remain vigilant in rural stretches.

9. Analysis of Communication Coverage

- Major highway stretches generally have good mobile coverage.
- Some rural segments may have spotty network service, particularly in hilly terrain or dense vegetation areas.

10. Estimated Emergency Response Times

- **Urban Areas:** Services can be expected within 30-45 minutes.
- **Rural Segments:** Response times may extend up to 1.5 hours due to distance and road access challenges.

11. Overall Summary of Risk Assessment

The Sangagiri to Karur route is generally navigable and safe for driving with proper precautions. The key risks include weather-related issues, particularly during the monsoon, and potential traffic congestion in urban areas. While infrastructure is good on national highways, attention is required on smaller roads. Emergency preparedness should focus on effective communication and understanding of local regulations, with emphasis on avoiding heavily congested times and routes. Environmental considerations and ensuring proper communication capabilities are critical for reducing risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.96304, 78.06131	15 KM/Hr
2	Roundabout	High	10.95934, 78.07586	15 KM/Hr
3	Roundabout	High	10.95613, 78.08207	15 KM/Hr
4	Roundabout	High	10.95024, 78.08300	15 KM/Hr
5	Turn	Medium	11.43822, 77.87348	30 KM/Hr
6	Turn	High	11.43968, 77.87345	15 KM/Hr
7	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
8	Turn	High	11.37862, 77.89488	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.37850, 77.89453	15 KM/Hr
10	Turn	Medium	11.16764, 78.01565	30 KM/Hr
11	Turn	High	11.15573, 78.02176	15 KM/Hr
12	Turn	High	11.14545, 78.02208	15 KM/Hr
13	Turn	High	11.14480, 78.02291	15 KM/Hr
14	Turn	High	10.95927, 78.07638	15 KM/Hr
15	Turn	Medium	10.95601, 78.07659	30 KM/Hr
16	Turn	High	10.95586, 78.07671	15 KM/Hr
17	Turn	Medium	10.95544, 78.08199	30 KM/Hr
18	Turn	Medium	10.95071, 78.08817	30 KM/Hr
19	Turn	Medium	10.95041, 78.08842	30 KM/Hr
20	Turn	High	10.94503, 78.11157	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
19	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
20	clinic	Senthil Nursing Home, Karur	10.9620158, 78.0643985	30 km/h	Medium
21	hospital	Apollo Hospital	10.96044, 78.064468	30 km/h	Medium
22	hospital	Murugesan Hospital	10.9619995, 78.0643134	30 km/h	Medium
23	hospital	Senthil's Emergency Care	10.9619826, 78.0643823	30 km/h	Medium
24	hospital	Kumaran Hospital	10.9619213, 78.0644924	30 km/h	Medium
25	hospital	Vasan Eye Care Hospital, Karur	10.9608117, 78.0665959	30 km/h	Medium
26	clinic	V.C Nursing Home	10.9605156, 78.0663214	30 km/h	Medium
27	hospital	Amaravathi Hospital	10.9602192, 78.0684534	30 km/h	Medium
28	hospital	Karuvurar Siddha Hospital	10.9617419, 78.0689621	30 km/h	Medium
29	hospital	Arasan Eye Hospital	10.961065, 78.067647	30 km/h	Medium
30	hospital	K R G Hospital	10.9633415, 78.0684828	30 km/h	Medium
31	hospital	Loganathan Dental Care Hospital	10.962963, 78.068556	30 km/h	Medium
32	hospital	Gugan Hospital	10.9601178, 78.0691481	30 km/h	Medium
33	hospital	Poongothai Hospital	10.9619987, 78.0690593	30 km/h	Medium
34	hospital	Ranjith Child Hospital	10.9622369, 78.0690826	30 km/h	Medium
35	clinic	Renu Nursing Home	10.9594975, 78.0704542	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	hospital	Nesi Eye Care Hospital	10.9612243, 78.0704182	30 km/h	Medium
37	hospital	S.G. Hospital	10.9610605, 78.0715124	30 km/h	Medium
38	hospital	Ammaiyappa Hospital	10.95901, 78.072469	30 km/h	Medium
39	clinic	Rainbow Baby Clinic	10.9627709, 78.0718342	30 km/h	Medium
40	clinic	Dr. Prabakaran ENT Clinic	10.9595218, 78.0735044	30 km/h	Medium
41	hospital	Kabila Kannan Hospital	10.959361, 78.072534	30 km/h	Medium
42	hospital	Krg Hospitals	10.9620644, 78.07212	30 km/h	Medium
43	hospital	Parthasarathy's Dental Hospital	10.9615774, 78.0716249	30 km/h	Medium
44	hospital	Neethu Neuro Hospital	10.958315, 78.0734656	30 km/h	Medium
45	clinic	Surya Clinic	10.9593375, 78.0736419	30 km/h	Medium
46	hospital	Niha Multi Speciality Hospitals	10.9611515, 78.0715536	30 km/h	Medium
47	clinic	Swaasthya Ayurveda Clinic	10.9587162, 78.0726022	30 km/h	Medium
48	hospital	Renuga Hospital	10.9595907, 78.0733125	30 km/h	Medium
49	hospital	Dhanvanthri Nilayam	10.9616527, 78.0735138	30 km/h	Medium
50	hospital	Velan Eye Hospital, Karur	10.958975, 78.0739935	30 km/h	Medium
51	hospital	Aarthy Eye Hospital	10.9607775, 78.0741034	30 km/h	Medium
52	hospital	Karur Dental and Cosmetology Hospital	10.9617053, 78.0752566	30 km/h	Medium
53	clinic	KB Kavitha Skin Care Centre	10.9616099, 78.0766208	30 km/h	Medium
54	clinic	K.S Skin Clinic	10.9617018, 78.0756076	30 km/h	Medium
55	clinic	Sri Vinayakah Kids and ENT Clinic	10.9618112, 78.0763833	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
56	hospital	Dr. Karunanidhi Baby Care Hospital	10.962121, 78.0741953	30 km/h	Medium
57	clinic	Krishna Neuro Centre	10.9614439, 78.0737188	30 km/h	Medium
58	clinic	Dr. Thirumalaisamy ENT Clinic	10.9620139, 78.0756801	30 km/h	Medium
59	hospital	Skin Trust Hospital	10.9615302, 78.0767588	30 km/h	Medium
60	hospital	Akhila Children's Hospital	10.9585894, 78.0745699	30 km/h	Medium
61	hospital	Balaji Hospital	10.9613902, 78.076345	30 km/h	Medium
62	clinic	Karur Neuro Centre	10.9587316, 78.0741106	30 km/h	Medium
63	clinic	Sankar Gastro Care	10.9617565, 78.0764935	30 km/h	Medium
64	clinic	Aadhavan Diet Clinic	10.9622519, 78.0744953	30 km/h	Medium
65	hospital	Jeyamani Plastic Surgery Centre	10.9615223, 78.0737932	30 km/h	Medium
66	clinic	Dr. Saravanan Sarswathi Homeo Clinic	10.9618052, 78.0746791	30 km/h	Medium
67	clinic	Nivi's Smile Care	10.9591388, 78.0741614	30 km/h	Medium
68	clinic	Rasammal Baby Clinic	10.9620194, 78.0744867	30 km/h	Medium
69	hospital	bhawa medical	10.9589205, 78.0761443	30 km/h	Medium
70	clinic	Karur Nursing Home	10.9615319, 78.0769278	30 km/h	Medium
71	hospital	G.V Hospital	10.9607995, 78.0773831	30 km/h	Medium
72	clinic	Dr. Rajendran Clinic	10.9619729, 78.0769794	30 km/h	Medium
73	hospital	K G Hospital, Karur	10.9555161, 78.0771773	30 km/h	Medium
74	clinic	Siddha Skin Clinic	10.9564121, 78.0799866	30 km/h	Medium
75	hospital	Skin Hospital	10.9560809, 78.0799651	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
76	hospital	Raja Psychiatric Hospital	10.956568, 78.079981	30 km/h	Medium
77	hospital	Jai Children Hospital	10.956354, 78.080021	30 km/h	Medium
78	clinic	Meenakshi Skin Clinic	10.956036, 78.0804762	30 km/h	Medium
79	clinic	Sitharth Skin Clinic	10.9563547, 78.0805722	30 km/h	Medium
80	clinic	Sri Anjana Nursing Home	10.9558611, 78.0816284	30 km/h	Medium
81	hospital	Abirami Maternity Centre	10.9559564, 78.0809939	30 km/h	Medium
82	hospital	Manimekalai Hospital	10.955985, 78.081226	30 km/h	Medium
83	clinic	Preethi Clinic	10.95625, 78.0811168	30 km/h	Medium
84	hospital	Ananya Ortho Care Hospital	10.956319, 78.081435	30 km/h	Medium
85	hospital	K.V.S Hospital	10.9560191, 78.0809837	30 km/h	Medium
87	pharmacy	thulasi	10.9501692, 78.0885672	30 km/h	Medium
89	clinic	K.K. Clinic	10.9472202, 78.1034518	30 km/h	Medium
90	hospital	Vidiyal Hospital	10.9479113, 78.1004165	30 km/h	Medium
91	clinic	Vignesh Clinic	10.9468071, 78.1051861	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
86	school	little angles	10.9556249, 78.0847999	30 km/h	Medium
88	school	St Theresa's Girls Higher Secondary School	10.9490521, 78.0949561	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

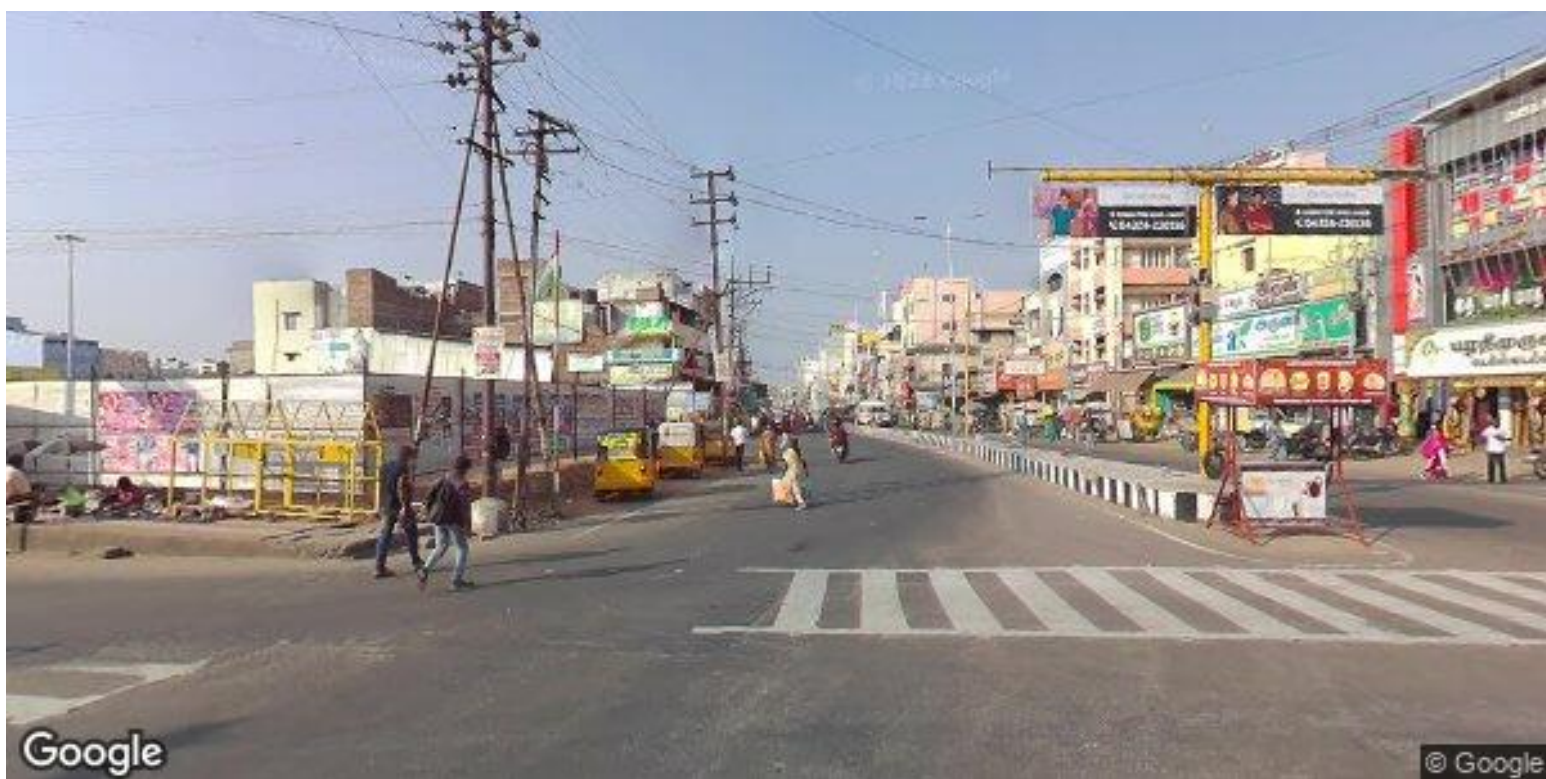


Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96304, 78.06131



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95934, 78.07586



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95613, 78.08207



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95024, 78.08300



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95927, 78.07638



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95601, 78.07659



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95586, 78.07671



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95544, 78.08199



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95071, 78.08817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95041, 78.08842



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.94503, 78.11157
